



STEM SMART Double Maths 22 - Vector Equations of Motion >

Acceleration f(t) 3ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

A cyclist travels along a straight road. Her velocity $v \text{ m s}^{-1}$, at time t seconds after starting from a point O, is given by

$$v = 2 \text{ for } 0 \leq t \leq 10$$

$$v = 0.03t^2 - 0.3t + 2 \text{ for } t \geq 10$$

Part A

Displacement at $t = 10$

Find the displacement of the cyclist from O when $t = 10$ s.

Part B

Expression for displacement

Find an expression for the displacement of the cyclist from O as a function of time for $t \geq 10$ s. Give your answer using fractions, not decimals.

The following symbols may be useful: t

Part C
Time

Find the time when the acceleration of the cyclist is 0.6 m s^{-2} .

Part D
Displacement

Find the displacement of the cyclist from O when her acceleration is 0.6 m s^{-2} .

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Calculus and Vectors 1ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

A particle P of mass 0.2 kg moves on a smooth horizontal plane. Initially it is projected with velocity 0.8 m s^{-1} from a fixed point O towards another fixed point A . At time $t\text{ s}$ after projection, P is $x\text{ m}$ from O and is moving with velocity $v\text{ m s}^{-1}$, with the direction OA being positive. A force of $(1.5t - 1)\text{ N}$ acts on P in the direction parallel to OA .

Part A

Expression for v

Find an expression for v in terms of t .

(Use fractions rather than decimals when entering your answer.)

The following symbols may be useful: t , v

Part B

Time when $v = 0.8\text{ m s}^{-1}$

Find the time (in seconds) when the velocity of P is next 0.8 m s^{-1} .

Give your answer to 3 sf.

Part C**Times through O**

Find the first time when P subsequently passes through O.

Find the second time when P subsequently passes through O.

Part D**Distance in third second**

Find the distance P travels in the third second of its motion.

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Kinematics & Calculus

Subject & topics: Maths Stage & difficulty: A Level C2

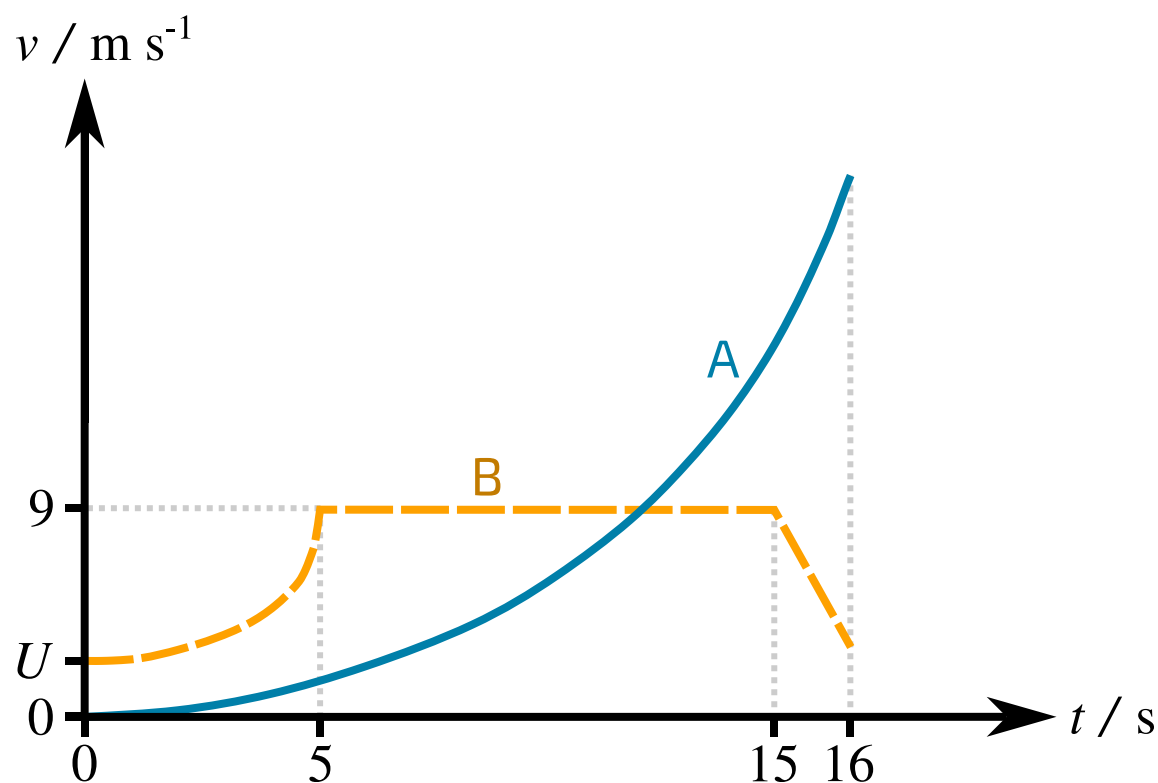


Figure 1: Velocity-time graph of the motion of two particles A and B along the same straight line.

The diagram shows the (t, v) graphs for two particles A and B which move on the same straight line. The units of v and t are m s^{-1} and s respectively. Both particles are at the point S on the line when $t = 0$. The particle A is initially at rest, and moves with acceleration $0.18t \text{ m s}^{-2}$ until the two particles collide when $t = 16 \text{ s}$. The initial velocity of B is $U \text{ m s}^{-1}$ and B has variable acceleration for the first five seconds of its motion. For the next ten seconds of its motion B has a constant velocity of 9 m s^{-1} ; finally B moves with constant deceleration for one second before it collides with A.

Part A

t for same velocity

Calculate the value of t at which the two particles have the same velocity.

Part B

Calculate U

For $0 \leq t \leq 5$ the distance of B from S is $(Ut + 0.08t^3)$ m.

Calculate U .

Part C

Distance from S

Calculate how far B is from S when $t = 5$ s.

Part D

 v_B when $t = 16$ s

Calculate the velocity of B when $t = 16$ s.

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Vectors & Calculus 2i

Subject & topics: Maths **Stage & difficulty:** A Level P3

A projectile has velocity $\begin{pmatrix} A \\ 5 - gt \end{pmatrix} \text{ m s}^{-1}$, where A is a constant.

Part A

Displacement

The particle is at $\begin{pmatrix} 5 \\ 10 \end{pmatrix}$ when $t = 0$.

Find an expression for the x -component of the particle's displacement, in metres, as a function of t .

The following symbols may be useful: A , g , t

Find an expression for the y -component of the particle's displacement, in metres, as a function of t .

The following symbols may be useful: A , g , t

Part B

Force

Find an expression for the force on the particle, given that it has mass m kg. Give your answer in the form $a\underline{i} + b\underline{j}$ where $\underline{i}$ and $\underline{j}$ are unit vectors in the x and y directions respectively.

The following symbols may be useful: A , g , i , j , m

Part C

Value of A

The projectile hits a target at the coordinates $\begin{pmatrix} 20 \\ 0 \end{pmatrix}$.

What is the value of A ? Give your answer to 2 significant figures. In your calculation, use the approximation $g \approx 10 \text{ m s}^{-2}$ and assume that the target is hit at $t > 0$.

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Vectors & Calculus 1i

Subject & topics: Maths **Stage & difficulty:** A Level P3

A planet moves through space. The force on the planet is given by

$$\underline{\boldsymbol{F}} = \begin{pmatrix} -mAB^2 \cos Bt \\ -mAB^2 \sin Bt \end{pmatrix}$$

where A and B are numerical constants and m is the mass of the planet.

Part A

Velocity

Given that the velocity of the planet when $t = 0$ is $\begin{pmatrix} 0 \\ AB \end{pmatrix}$.

Find an expression for the x -component of the velocity of the planet as a function of time.

The following symbols may be useful: A , B , $\cos()$, $\sin()$, t , $\tan()$

Find an expression for the y -component of the velocity of the planet as a function of time.

The following symbols may be useful: A , B , $\cos()$, $\sin()$, t , $\tan()$

Part B**Displacement**

Given that the displacement of the planet when $t = 0$ is $\begin{pmatrix} A \\ 0 \end{pmatrix}$.

Find an expression for the x -component of the displacement of the planet as a function of time.

The following symbols may be useful: A , B , $\cos()$, $\sin()$, t , $\tan()$

Find an expression for the y -component of the displacement of the planet as a function of time.

The following symbols may be useful: A , B , $\cos()$, $\sin()$, t , $\tan()$

Part C**Modulus**

Find an expression for the modulus of the displacement. Simplify your answer as far as possible.

The following symbols may be useful: A , B , $\cos()$, $\sin()$, t , $\tan()$

Part D**Shape of path**

What is the shape of the path that the planet follows?

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Vector Equations of Motion 1

Subject & topics: Maths | Geometry | Vectors

Stage & difficulty: A Level P3

This question looks at three different uses of calculus in vector problems.

Part A

Integrating to find particle displacement

A particle moves in the x - y plane with velocity $\underline{v} = \begin{pmatrix} 2te^{-2t^2} \\ 3te^{-4t^2} \end{pmatrix}$. Find an expression for the displacement of the particle at time t , given that the particle is at the origin when $t = 0$.

Enter an expression for the x -component of the displacement.

The following symbols may be useful: e, t

Enter an expression for the y -component of the displacement.

The following symbols may be useful: t

Part B

Finding a maximum speed

At a time t s a particle moves in the x - y plane with velocity $\underline{v} = \begin{pmatrix} 2te^{-2t^2} \\ 3 \end{pmatrix} \text{ m s}^{-1}$. What is the maximum speed of the particle? Give your answer as an expression in terms of e.

The following symbols may be useful: e

Part C

Distance of closest approach to the origin

The displacement of a particle is given by the expression $\underline{s} = \begin{pmatrix} e^{3t} \\ e^{6t} - 5 \end{pmatrix}$. Find the shortest distance between the particle and the origin during the particle's motion. Give your answer in the form $\frac{\sqrt{a}}{2}$.

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Projectiles: Trajectories 3i

Subject & topics: Maths Stage & difficulty: A Level P2

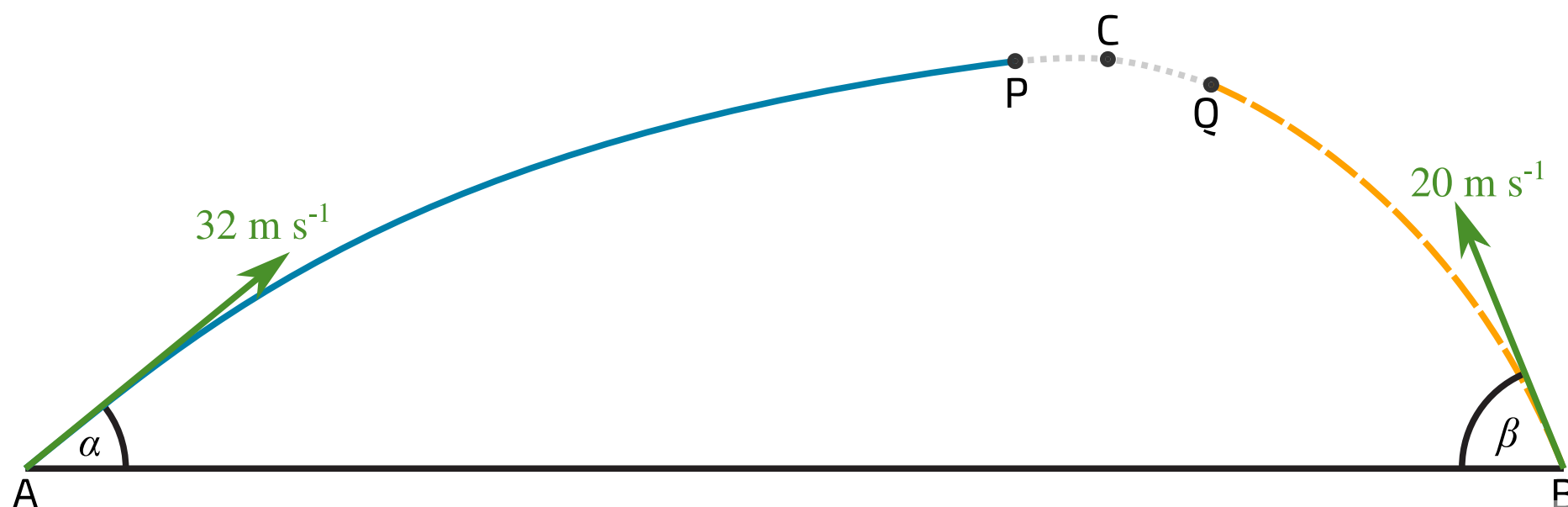


Figure 1: The trajectories of particles P and Q.

A particle P is projected with speed 32 m s^{-1} at an angle of elevation α , where $\sin \alpha = \frac{3}{5}$, from a point A on horizontal ground. At the same instant a particle Q is projected with speed 20 m s^{-1} at an angle of elevation β , where $\sin \beta = \frac{24}{25}$, from a point B on the same horizontal ground. The particles move freely under gravity in the same vertical plane and collide with each other at the point C at the instant when they are travelling horizontally.

Part A

Velocities of P and Q

Express the initial velocities of P and Q in vector form using the unit vectors $\underline{i}$ and $\underline{j}$, where $\underline{i}$ is a unit vector in the direction of $\overrightarrow{AB}$ and $\underline{j}$ is a unit vector vertically upwards.

$$\underline{u}_P = \frac{1}{5} \left(\quad \underline{i} + \quad \underline{j} \right) \text{ ms}^{-1}$$

$$\underline{u}_Q = \frac{1}{5} \left(\quad \underline{i} + \quad \underline{j} \right) \text{ ms}^{-1}$$

Part B

Height of C

Calculate the height of C above the ground. Give your answer to 3 significant figures.

Part C

Time in air

Find the time, t , between projection and collision. Give your answer to 3 significant figures.

Part D

Distance AB

Calculate the distance AB. Give your answer to 3 significant figures.

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Ships at Sea

Subject & topics: Maths

Stage & difficulty: A Level C2

Part A

Unit vector

Find the unit vector in the same direction as $\underline{p} = 6\underline{i} - 3\underline{j} + 2\underline{k}$. Write your answer in terms of the unit vectors $\underline{i}$, $\underline{j}$ and $\underline{k}$.

The following symbols may be useful: $\underline{i}$, $\underline{j}$, $\underline{k}$

Part B

Vector parallel to $\underline{p}$

Hence, find a vector of magnitude 4 parallel to $\underline{p}$. Write your answer in terms of the unit vectors $\underline{i}$, $\underline{j}$ and $\underline{k}$.

The following symbols may be useful: $\underline{i}$, $\underline{j}$, $\underline{k}$

Part C

S and T

In an experiment two ships, S and T, move in a calm sea.

In a simple model, S and T are treated as objects whose dimensions are negligible. Which single word describes an object for which such a modelling assumption is made?

Part D

Position of S

S starts from the origin and moves with constant velocity $\underline{v}_1 = 4\underline{i} - 3\underline{j}$, where $\underline{v}_1$ is measured in km h^{-1} .

Write down the position vector of S in terms of t using the unit vectors $\underline{i}$, $\underline{j}$ and $\underline{k}$.

The following symbols may be useful: i, j, k, t

Part E

Calm sea assumption

In the experiment the two ships, S and T, move in a calm sea.

Using the information given, justify limiting the velocity vectors of S and T to two dimensions.

The sea is described as calm so we assume the surface to be a so we only require perpendicular directions to describe the motion rather than .

Items:

point

line

plane

space

1

2

3

Part F

Do they meet?

T starts from the point with position vector $(3\underline{i} - 5\underline{j})$ and moves with constant velocity $\underline{v}_2 = (\underline{i} + 4\underline{j})$ where $\underline{v}_2$ is measured in km h^{-1} .

Do S and T ever meet? If so, find the time in seconds. If not, enter in 0.

Part G

Minimum distance

Find the minimum distance between S and T. Give your answer using exact fractions and surds.

Part H

Finding y

A second experiment is undertaken. The behavior of S is unchanged. T starts from the same point as before, but this time it is the intention that S and T meet. The required constant velocity to set for T is of the form $(\underline{i} + y\underline{j})$.

Find the value of y .

Part I

Percentage accuracy

The ships are each approximately 40 m long. Estimate the maximum percentage error in the calculation of the time it takes for the ships to meet due to using the assumption described in [Part B](#). Give your answer to 1 significant figure.

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